



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA -2
Academic year	I – III Semester(2025-26)
Faculty Name	Mohammed Razia Alangir Banu
Student Name	B.Deepak Goud
Roll Number	2520080035
Branch	AI&DS
Course Outcome	2

Case Study Topic : FoodDash Hyderabad — delivery-fee heatmap optimization

Word Done Summary:

The FoodDash Hyderabad Delivery-Fee Heatmap Optimization System was developed to analyze food delivery data and optimize delivery charges across different regions of Hyderabad. The project involved collecting order, distance, traffic, and demand information, generating heatmaps to visualize fee patterns, identifying high-cost delivery zones, and implementing dynamic pricing strategies. The system helps improve delivery efficiency, reduce operational costs, ensure fair pricing for customers, and support better decision-making through data-driven insights and visual analytics.

Implementation Details:

1. Collected and processed delivery, traffic, and demand data from different regions.
2. Generated heatmaps to visualize delivery fee patterns across Hyderabad.
3. Applied optimization algorithms to calculate efficient delivery charges.
4. Developed a dashboard to monitor fees, demand, and delivery performance.

Result:

The FoodDash Hyderabad Delivery-Fee Heatmap Optimization System successfully identified delivery fee patterns across different regions and improved pricing accuracy through data-driven analysis. The generated heatmaps provided clear insights into high-demand and high-cost delivery zones, enabling better fee optimization and resource allocation. As a result, the system helped reduce operational inefficiencies, improve delivery planning, ensure fair pricing for customers, and support informed business decision-making.

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● b.deepak@BDeepaks-MacBook-Air DSA project % javac SegTreeLazy.java
● b.deepak@BDeepaks-MacBook-Air DSA project % java SegTreeLazy
  Max in [0,15] = 9
  Max in [3,10] = 11
○ b.deepak@BDeepaks-MacBook-Air DSA project % █
```

GitHub Link:

<https://github.com/bdeepakg-boop/DSA-Project/upload/main>

Student Signature	Faculty Signature